

DATA SCIENCE RESEARCH CAPSTONE

Predictive Analytics for Daily Demand

A Data Science Research Project

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Research Focus

This study investigates whether historical demand patterns, calendar characteristics, marketing events, and holidays can be used to predict daily demand using statistical analysis and machine learning.

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Abstract

Demand forecasting is an important component of data-driven business decision-making because accurate estimates of future demand can support inventory planning, workforce allocation, marketing activities, and operational planning. This research investigates whether daily demand can be predicted using historical demand-related information, calendar characteristics, marketing events, and holidays.

The study uses a dataset containing 30 daily observations. The data was examined for missing values and duplicate records, and additional calendar features were created from the date variable. Exploratory data analysis was performed to identify demand patterns and relationships among the available variables. An independent two-sample t-test was also conducted to examine whether demand differed between marketing-event and non-marketing-event observations.

Two supervised machine learning models, Linear Regression and Random Forest Regression, were developed and evaluated using a chronological 80/20 train-test split. Model performance was measured using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R-squared (R^2).

On the test set, Linear Regression produced an MAE of 2.82, an RMSE of 3.15, and an R^2 of 0.9143. Random Forest produced an MAE of 5.45, an RMSE of 6.22, and an R^2 of 0.6658. The statistical analysis produced a t-statistic of 4.2128 and a p-value of 0.0014, indicating a statistically significant difference in mean demand between marketing-event and non-event observations at the 5% significance level.

Because the dataset contains only 30 observations and the test set contains six observations, the findings should be considered preliminary. Nevertheless, the study demonstrates a complete data science workflow from data preparation and statistical analysis through predictive modeling and interpretation.

1. Introduction

Organizations increasingly use data science to transform historical observations into information that supports planning and decision-making. One practical application is demand prediction. When a business can estimate demand for upcoming periods, it can better coordinate inventory, staffing, procurement, promotions, and other operational activities.

Demand is rarely determined by a single factor. Calendar effects, holidays, promotional events, seasonality, customer behavior, and other external variables can contribute to changes in demand. Predictive analytics provides a structured way to evaluate whether available variables contain useful information for estimating the target variable.

This capstone applies a complete predictive analytics workflow to a small daily-demand dataset. The workflow begins with data-quality inspection and feature engineering, continues through exploratory and statistical analysis, and concludes with machine learning models and performance evaluation.

1.1 Research Question

Can historical demand patterns, marketing events, holidays, and calendar-based characteristics be used to predict daily demand accurately?

1.2 Research Objectives

1. Analyze daily demand patterns over time.
2. Examine the relationship between demand, marketing events, and holidays.
3. Engineer useful calendar-based predictive features.
4. Develop machine learning models for daily demand prediction.
5. Compare model performance using MAE, RMSE, and R^2 .
6. Translate the analytical findings into practical business implications.

1.3 Research Hypotheses

H₀: Calendar characteristics and marketing/holiday events do not provide useful information for predicting daily demand.

H_a: Calendar characteristics and marketing/holiday events provide useful information for predicting daily demand.

2. Literature Background

Demand forecasting is a well-established application of statistical modeling and machine learning. Traditional forecasting methods often emphasize historical patterns, trend, seasonality, and time dependence. More recent predictive analytics workflows also incorporate explanatory variables and machine learning algorithms.

Linear regression is one of the most widely used supervised learning approaches for continuous targets. It estimates a linear relationship between explanatory variables and the response variable. Its advantages include interpretability, computational simplicity, and a direct connection between model coefficients and predictors.

Tree-based ensemble methods such as Random Forest can represent nonlinear relationships and interactions without requiring the analyst to specify those relationships in advance. Random Forest combines many decision trees and averages their predictions, which can provide flexible predictive behavior.

Model evaluation is essential because a model that fits historical observations well may not necessarily generalize to unseen observations. MAE summarizes the average absolute prediction error, RMSE gives greater weight to larger errors, and R^2 describes the proportion of target variation explained by the model under the evaluation setup.

For time-ordered data, the ordering of observations matters. A chronological train-test split helps avoid a situation in which information from future observations is used to train a model that is then evaluated on earlier observations. This capstone therefore uses the first 80% of the observations for training and the final 20% for testing.

2.1 Relevance to Business Analytics

Forecasting methods can support practical questions such as how much inventory to prepare, when staffing requirements may increase, and whether promotional periods are associated with changes in demand. However, the quality of these decisions depends on the quantity and quality of historical data and the completeness of the predictor set.

3. Data and Methodology

3.1 Dataset Description

The dataset contains 30 daily observations covering June 2026. The original variables are date, demand, marketing_event, and holiday. The demand variable is the continuous target used in the predictive modeling stage. Marketing event and holiday are binary indicators.

Variable	Type	Role	Description
date	Date	Temporal feature	Calendar date of observation
demand	Integer	Target	Daily demand value
marketing_event	Binary	Predictor	Indicator for a marketing event
holiday	Binary	Predictor	Indicator for a holiday

3.2 Data Quality

The dataset was checked for missing values and duplicate rows. The analysis found **0 missing values** and **0 duplicate rows**. The demand variable had a mean of 142.10 and a standard deviation of 13.84, with observed values ranging from 118 to 174.

3.3 Feature Engineering

The date column was converted to datetime format and sorted chronologically. Three calendar-derived features were created: day of month, day of week, and week of month. Together with the marketing-event and holiday indicators, these variables formed the predictor set for the regression models.

3.4 Train-Test Design

A chronological 80/20 split was applied. The first 24 observations were used for model training and the final 6 observations were held out for testing. This design preserves temporal ordering and is more appropriate for this small time-ordered dataset than a random split that could mix earlier and later observations.

3.5 Software and Tools

The analysis was implemented in Python using Google Colab. The workflow used Pandas and NumPy for data handling, Matplotlib and Seaborn for visualization, SciPy for statistical testing, and scikit-learn for machine learning and model evaluation.

4. Exploratory Data Analysis

Exploratory data analysis was conducted before predictive modeling to understand the structure of the data and identify relationships that could inform model development. The notebook includes a time-series plot of daily demand, a demand distribution plot, boxplots comparing demand across marketing-event and holiday indicators, and a numerical correlation matrix.

4.1 Demand Distribution

The observed demand values range from 118 to 174, with a mean of 142.10 and a standard deviation of 13.84. The median demand is 142. These summary statistics indicate that demand varies meaningfully around its central level within the observed 30-day period.

4.2 Marketing Events and Holidays

Marketing events and holidays were represented as binary variables. Grouped summary statistics and boxplots were used to compare demand distributions for event versus non-event days and holiday versus non-holiday days. These comparisons provide a direct descriptive view of whether demand differs across the available categorical conditions.

4.3 Correlation Analysis

A correlation matrix was generated for demand and the engineered numerical features. Correlation analysis is descriptive rather than causal: a correlation does not establish that a variable causes changes in demand. The matrix was therefore used as an exploratory tool alongside the statistical test and predictive models.

Notebook evidence: The full EDA figures and tables are retained in the accompanying Google Colab notebook, allowing the visual results to be reproduced from the code.

5. Statistical Analysis

An independent two-sample t-test was conducted to evaluate whether average demand differed between observations with a marketing event and observations without a marketing event. The significance threshold was set at $\alpha = 0.05$.

5.1 Hypotheses

Null hypothesis (H_0): The mean demand is the same for marketing-event and non-marketing-event observations.

Alternative hypothesis (H_A): The mean demand differs between marketing-event and non-marketing-event observations.

5.2 Test Result

Statistic	Value
t-statistic	4.2128
p-value	0.0014
Significance level (α)	0.05
Decision rule	Reject H_0 when $p < 0.05$

The p-value of 0.0014 is below the 0.05 significance threshold. Under the stated hypothesis test, this provides statistical evidence of a difference in mean demand between the two groups in this sample. This result describes an association in the observed dataset and does not by itself establish that marketing events caused the difference.

Because the dataset contains only 30 observations, the test result should be interpreted with the sample size and available variables in mind. A larger dataset would provide a stronger basis for estimating the stability of the observed relationship.

6. Predictive Modeling

The predictive modeling stage evaluated two regression algorithms: Linear Regression and Random Forest Regression. Both models used `marketing_event`, `holiday`, `day`, `day_of_week`, and `week_of_month` as predictors.

6.1 Linear Regression

Linear Regression provides an interpretable baseline model by estimating a linear relationship between the predictor variables and daily demand. It was fitted using the first 24 chronological observations and evaluated on the final six observations.

6.2 Random Forest Regression

Random Forest Regression was included as a nonlinear ensemble comparison. The model used 200 trees with a fixed random state of 42. Its inclusion allows the experiment to compare a relatively simple linear model with a more flexible tree-based method.

6.3 Performance Metrics

MAE measures the average absolute difference between actual and predicted values. **RMSE** is the square root of the mean squared error and places relatively greater weight on larger errors. **R²** summarizes the proportion of variation in the target explained by the fitted model under the evaluation setup.

6.4 Model Results

Model	MAE	RMSE	R ²
Linear Regression	2.815506	3.149515	0.914344
Random Forest	5.450833	6.220987	0.665813

On the six-observation chronological test set, Linear Regression produced lower MAE and RMSE and a higher R² than Random Forest. This is an observed comparison for this dataset and split; it should not be interpreted as evidence that Linear Regression will always outperform Random Forest on other datasets or future periods.

7. Findings and Business Implications

7.1 Key Findings

1. The dataset was complete for the analyzed fields, with zero missing values and zero duplicate rows.
2. Demand had a mean of 142.10 and a standard deviation of 13.84 over the 30 observations.
3. The marketing-event t-test produced $t = 4.2128$ and $p = 0.0014$, which is statistically significant at $\alpha = 0.05$.
4. Linear Regression achieved $MAE = 2.815506$, $RMSE = 3.149515$, and $R^2 = 0.914344$ on the chronological test set.
5. Random Forest achieved $MAE = 5.450833$, $RMSE = 6.220987$, and $R^2 = 0.665813$ on the same test set.
6. The small sample size limits the stability and generalizability of these estimates.

7.2 Inventory Planning

Demand predictions can support inventory planning by providing an estimate of expected requirements. In a larger production system, forecasts could be integrated with inventory thresholds and replenishment schedules.

7.3 Marketing Planning

The statistical analysis found a difference in mean demand between marketing-event and non-event observations in this sample. Businesses could use larger datasets to investigate which types of events are associated with the strongest changes and whether those effects remain after controlling for seasonality, price, and other factors.

7.4 Workforce and Operations

Improved demand estimates can also inform staffing, procurement, logistics, and service capacity. The usefulness of such forecasts depends on regular retraining and monitoring as new demand observations become available.

7.5 Decision-Support Perspective

The model should be viewed as a decision-support prototype rather than an autonomous decision system. Human review and additional business information remain important when forecasts are used for operational planning.

8. Limitations

1. The dataset contains only 30 observations, which is small for predictive modeling.
2. Only six observations were reserved for testing, making evaluation sensitive to individual observations.
3. The available predictors do not include potentially important factors such as price, weather, inventory availability, product mix, or customer behavior.
4. The analysis evaluates two model families and does not include extensive hyperparameter optimization.
5. A single chronological holdout was used rather than repeated rolling-origin validation.
6. The observed statistical association between marketing events and demand should not be interpreted as proof of causality.
7. The results are specific to the dataset and observation period and should not automatically be generalized to other populations or future periods.

9. Future Research

Future work should begin by collecting a substantially larger historical dataset covering multiple months or years. This would allow the study to capture seasonal patterns and support more reliable validation.

Additional variables such as price, discount level, weather, product availability, promotion type, customer traffic, and inventory levels could be incorporated. These features could help separate calendar effects from other drivers of demand.

Methodologically, future research could compare additional algorithms and time-series models, apply hyperparameter tuning, and use rolling or walk-forward validation. Confidence intervals and uncertainty estimates could also be incorporated to make the forecasting output more useful for operational decision-making.

10. Conclusion

This capstone demonstrated an end-to-end data science workflow for daily demand prediction. The process included data-quality assessment, data processing, feature engineering, exploratory data analysis, statistical hypothesis testing, predictive modeling, and evaluation using standard regression metrics.

The independent two-sample t-test produced a t-statistic of 4.2128 and a p-value of 0.0014, providing statistical evidence of a difference in mean demand between marketing-event and non-event observations in the analyzed sample at the 5% significance level.

For predictive modeling, Linear Regression achieved MAE 2.815506, RMSE 3.149515, and R^2 0.914344, while Random Forest achieved MAE 5.450833, RMSE 6.220987, and R^2 0.665813 on the chronological test set. Thus, within this particular experimental setup, the linear model generated smaller test-set errors.

The principal contribution of the project is the demonstration of a reproducible predictive analytics pipeline. The results also illustrate why model performance must be interpreted together with dataset size, validation design, and the completeness of available explanatory variables.

A larger and richer dataset would be the logical next step toward developing a more reliable operational demand-forecasting system.

11. References

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7. The accompanying Google Colab notebook, *Data_Science_Research_Capstone.ipynb*, contains the executable analysis, visualizations, statistical calculations, and model evaluation.

Appendix: Reproducibility

The complete computational workflow is maintained in the accompanying Colab notebook. The notebook should be submitted alongside this PDF and the GitHub repository so that the mentor can inspect and reproduce the analysis.